

< Q1 >

Select Language : C (GCC 9.2.0)

```

1  #include <stdio.h>
2  int main()
3  {
4      int n, i;
5      int at[100], bt[100], ct[100], tat[100], wt[100];
6      float avg_tat = 0, avg_wt = 0;
7      printf("Enter number of processes: ");
8      scanf("%d", &n);
9      for(i = 0; i < n; i++)
10     {
11         printf("\nEnter Arrival Time and Burst Time: ");
12         scanf("%d %d", &at[i], &bt[i]);
13     }
14     ct[0] = at[0] + bt[0];

```



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Select Language : C (GCC 9.2.0)

```

15     for(i = 1; i < n; i++)
16     {
17         if(ct[i-1] < at[i])
18             ct[i] = at[i] + bt[i];
19         else
20             ct[i] = ct[i-1] + bt[i];
21     }
22     for(i = 0; i < n; i++)
23     {
24         tat[i] = ct[i] - at[i];
25         wt[i] = tat[i] - bt[i];
26         avg_tat += tat[i];
27         avg_wt += wt[i];
28     }

```

```

        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];
        avg_tat += tat[i];
        avg_wt += wt[i];
    }
    avg_tat /= n;
    avg_wt /= n;
    printf("\n\nAverage Turnaround Time: %.2f", avg_tat);
    printf("\nAverage Waiting Time: %.2f\n", avg_wt);
    return 0;
}

```

#### Sample Test Case

##### ▼ Test Case - 1

Input :

```

3
0 2
1 2
2 3

```

Expected Output :

```

Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:

```

```

Average Turnaround Time: 3.33
Average Waiting Time: 1.00

```

< Q1 >

Select Language : C (GCC 9.2.0) ▼

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i, j, time = 0, min, idx;
6     int at[100], bt[100], ct[100], tat[100], wt[100];
7     int completed[100] = {0};
8     float avg_tat = 0, avg_wt = 0;
9
10    printf("Enter number of processes: ");
11    scanf("%d", &n);
12
13    for(i = 0; i < n; i++)
14    {

```

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Select Language : C (GCC 9.2.0)



```
15     printf("\nEnter Arrival Time and Burst Time for P%d: ", i+1);
16     scanf("%d %d", &at[i], &bt[i]);
17 }
18
19 int done = 0;
20
21 while(done < n)
22 {
23     min = 9999;
24     idx = -1;
25
26     for(i = 0; i < n; i++)
27     {
28         if(at[i] <= time && completed[i] == 0 && bt[i] < min)
```

Select Language : C (GCC 9.2.0)



```
29     {
30         min = bt[i];
31         idx = i;
32     }
33 }
34
35 if(idx != -1)
36 {
37     time += bt[idx];
38     ct[idx] = time;
39     completed[idx] = 1;
40     done++;
41 }
42 else
```

Select Language : C (GCC 9.2.0)



```
43     {
44         time++;
45     }
46 }
47
48 for(i = 0; i < n; i++)
49 {
50     tat[i] = ct[i] - at[i];
51     wt[i] = tat[i] - bt[i];
52
53     avg_tat += tat[i];
54     avg_wt += wt[i];
55 }
56
57 avg_tat /= n;
58 avg_wt /= n;
59
60 printf("\n\nAverage Turnaround Time = %.2f\n", avg_tat);
61 printf("Average Waiting Time = %.2f\n", avg_wt);
62
63 return 0;
64 }
```

Sample Test Case

▼ Test Case - 1

Input :

```
4
0 6
0 8
0 7
0 3
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:
```

```
Average Turnaround Time = 13.00
Average Waiting Time = 7.00
```

< Q1 >

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2 int main() {
3     int n, i, j, time = 0, completed = 0;
4     int at[50], bt[50], pr[50];
5     int ct[50], tat[50], wt[50];
6     int done[50] = {0};
7     float avg_tat = 0, avg_wt = 0;
8     printf("Enter number of processes: ");
9     scanf("%d", &n);
10    for(i = 0; i < n; i++) {
11        printf("\nEnter AT, BT and Priority for P%d", i+1);
12        scanf("%d %d %d", &at[i], &bt[i], &pr[i]);
13    }
```

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```
14    while(completed < n) {
15        int idx = -1;
16        int highest = 9999;
17        for(i = 0; i < n; i++) {
18            if(at[i] <= time && done[i] == 0) {
19                if(pr[i] < highest) {
20                    highest = pr[i];
21                    idx = i;
22                }
23            }
24        }
25        if(idx != -1) {
26            time += bt[idx];
```

```
26         time += bt[idx];
27         ct[idx] = time;
28         tat[idx] = ct[idx] - at[idx];
29         wt[idx] = tat[idx] - bt[idx];
30         avg_tat += tat[idx];
31         avg_wt += wt[idx];
32         done[idx] = 1;
33         completed++;
34     }
35     else {
36         time++;
37     }
38 }
    }
}
avg_tat /= n;
avg_wt /= n;
printf("\n\nAverage Turnaround Time = %.2f\n", avg_tat);
printf("Average Waiting Time = %.2f\n", avg_wt);
return 0;
}
```

Test Case - 1

Input :

```
3
0 5 2
1 3 1
2 2 3
```

Expected Output :

```
Enter number of processes:
Enter AT, BT and Priority for P1:
Enter AT, BT and Priority for P2:
Enter AT, BT and Priority for P3:

Average Turnaround Time = 6.67
Average Waiting Time = 3.33
```

< Q1 >

Select Language : C (GCC 9.2.0)

```
1 #include <stdio.h>
2 int main() {
3     int n, tq, i, time = 0, completed = 0;
4     int at[50], bt[50], rt[50];
5     int ct[50], tat[50], wt[50];
6     float avg_tat = 0, avg_wt = 0;
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9     printf("\nEnter Time Quantum: ");
10    scanf("%d", &tq);
11    for(i = 0; i < n; i++) {
12        printf("\nEnter AT and BT for P%d: ", i+1);
13        scanf("%d %d", &at[i], &bt[i]);
14        rt[i] = bt[i];
15    }
16    while(completed < n) {
17        int executed = 0;
18        for(i = 0; i < n; i++) {
19            if(at[i] <= time && rt[i] > 0) {
20                executed = 1;
21                if(rt[i] > tq) {
22                    time += tq;
23                    rt[i] -= tq;
24                } else {
25                    time += rt[i];
26                    ct[i] = time;
```

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```
27         rt[i] = 0;
28         completed++;
29         tat[i] = ct[i] - at[i];
30         wt[i] = tat[i] - bt[i];
31         avg_tat += tat[i];
32         avg_wt += wt[i];
33     }
34 }
35 }
36 if(executed == 0)
37     time++;
38 }
39 avg_tat /= n;
avg_tat /= n;
avg_wt /= n;
printf("\n\nAverage Turnaround Time: %.2f\n", avg_tat);
printf("Average Waiting Time: %.2f\n", avg_wt);
return 0;
}
```

Sample Test Case

Test Case - 1

Input :

```
3
2
0 5
1 3
2 1
```

Expected Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```